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(54) **REVERSIBLE MIXING SHAFT AND ATTACHMENT FOR STAND MIXER**

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(57) **ABSTRACT**

(51) **Int. Cl.**

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A47J 43/08 (2006.01)

A stand mixer is provided having a motor, an attachment support coupled to the motor, a mixing shaft coupled to the motor, a retractable plunger, and a mixing attachment. The mixing shaft may be rotatable by the motor. The retractable plunger may be extendable between a first position into the mixing shaft and a second position extending from the mixing shaft. The mixing attachment may be removably attached to the mixing shaft and may define a shaft channel for sliding receipt of the mixing shaft. The mixing attachment may also define an aperture connected with the with the shaft channel and configured for selective receipt of the retractable plunger. The mixing shaft can be inserted into the shaft channel and may be moved to selectively lock the mixing attachment to the mixing shaft by receipt of the retractable plunger into the aperture of the mixing attachment.

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(2013.01); *A47J 43/082* (2013.01); *A47J*
2043/04454 (2013.01)

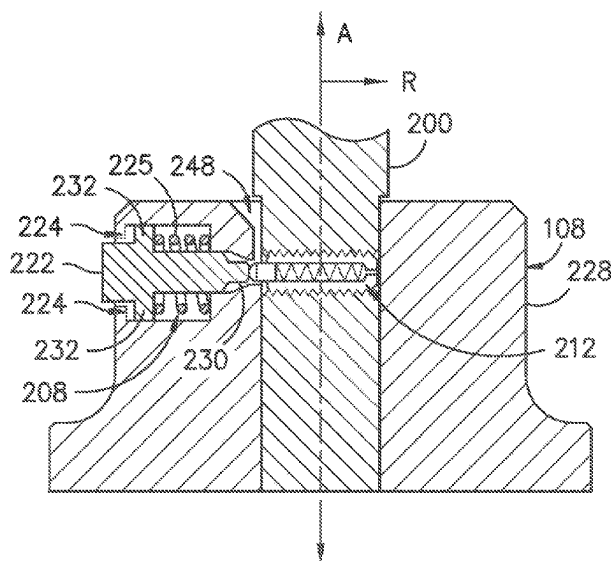
(58) **Field of Classification Search**

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USPC 366/197, 331, 343

See application file for complete search history.

18 Claims, 6 Drawing Sheets



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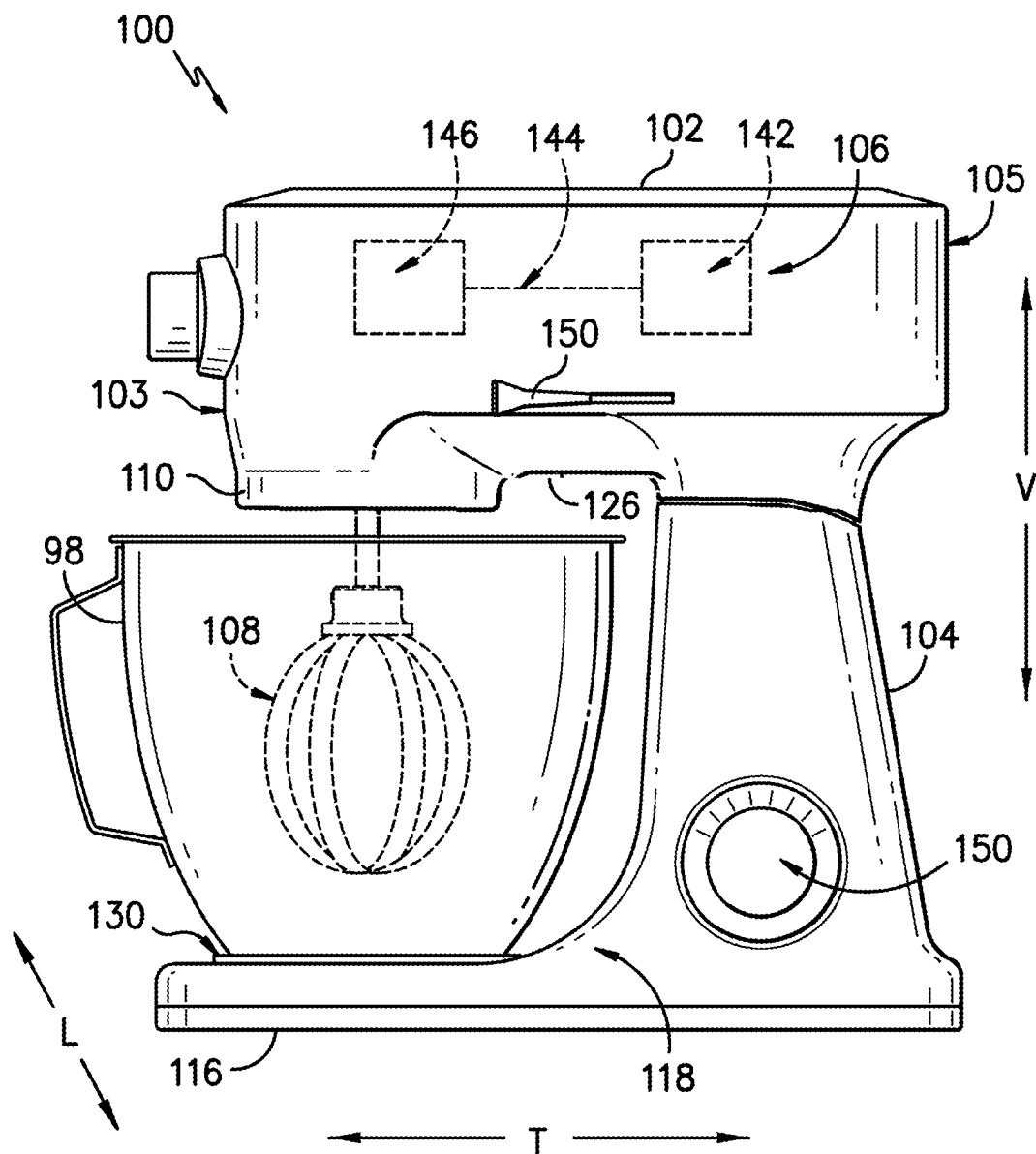
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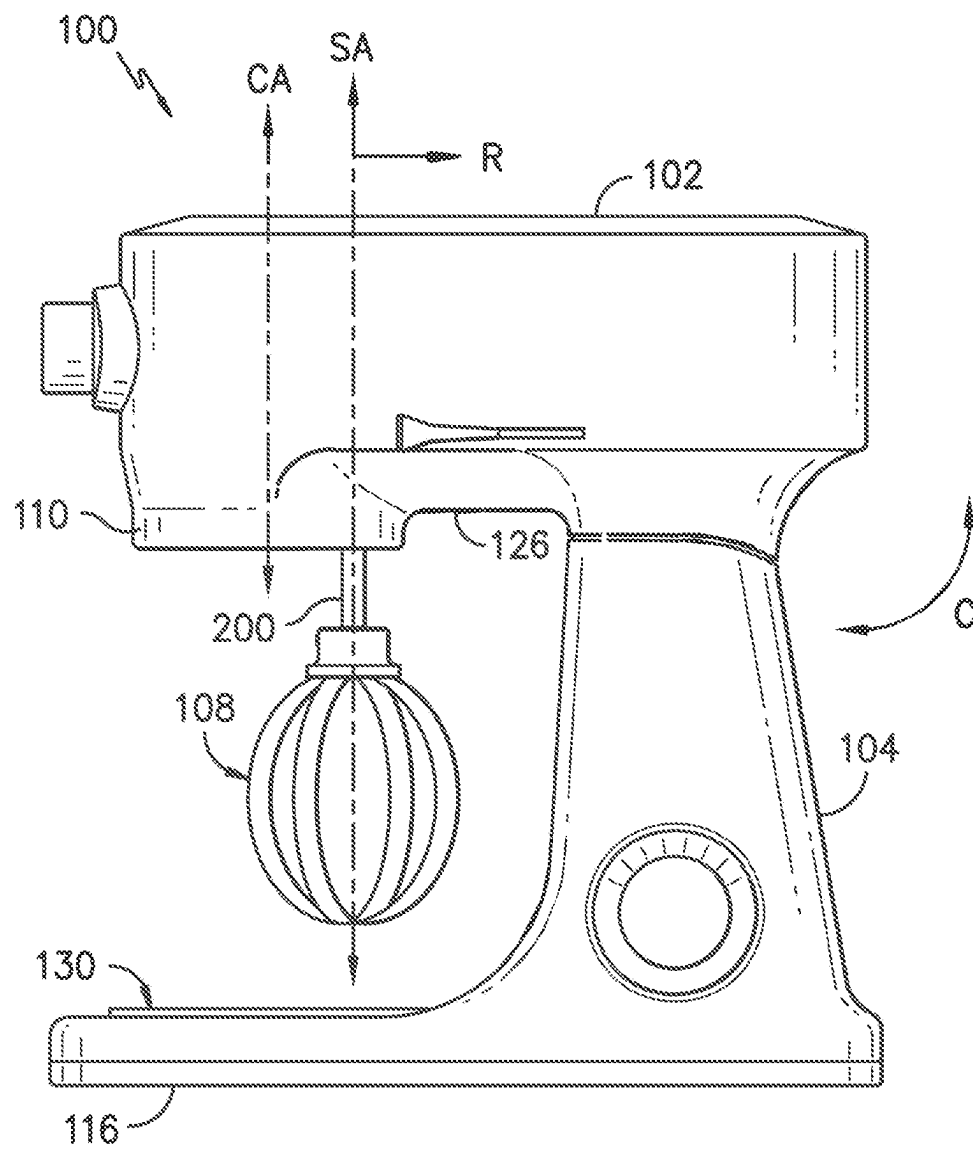
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*FIG. -1-*

*FIG. -2-*

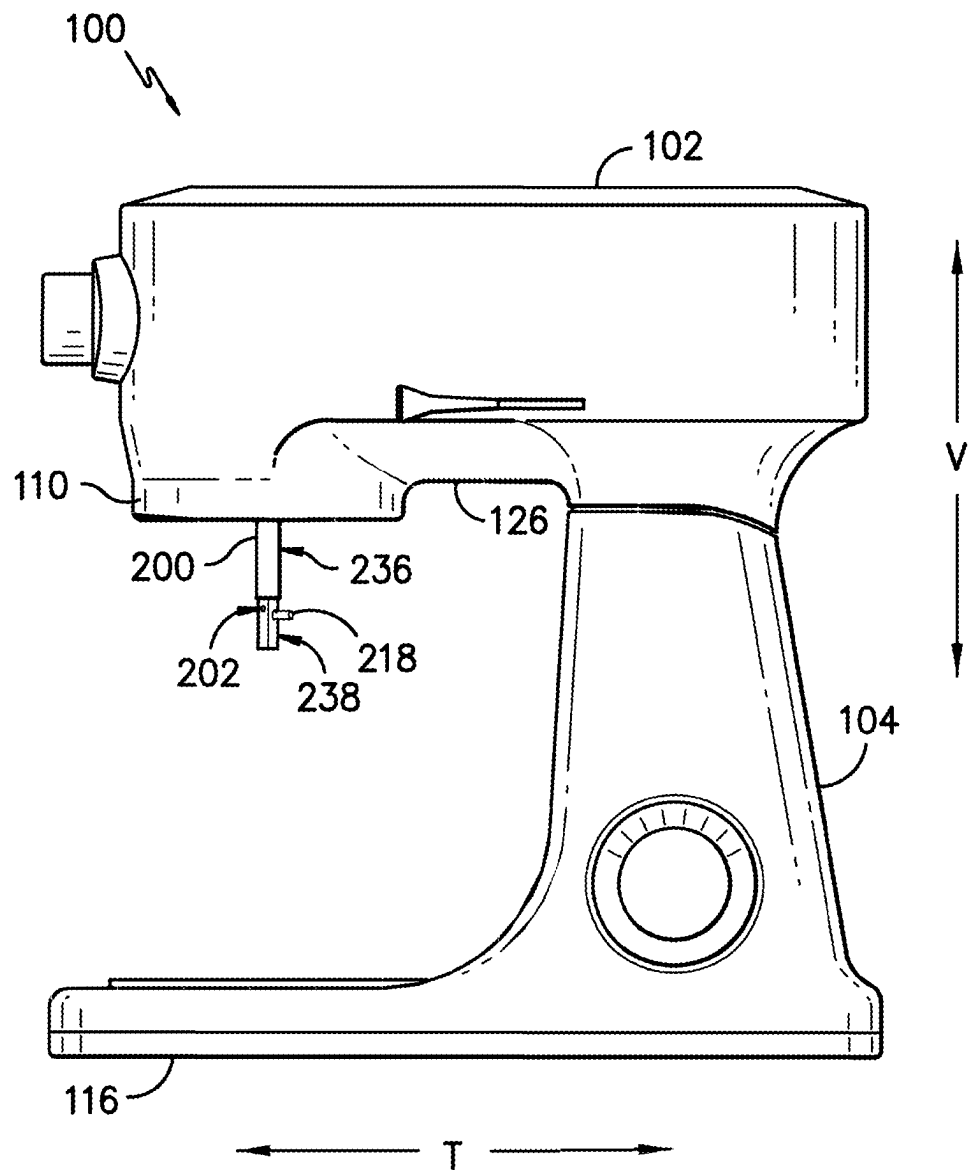
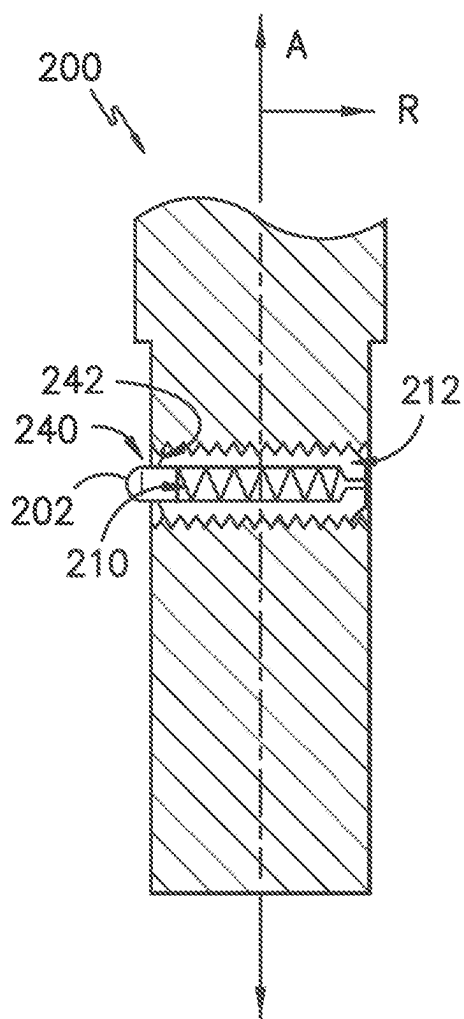
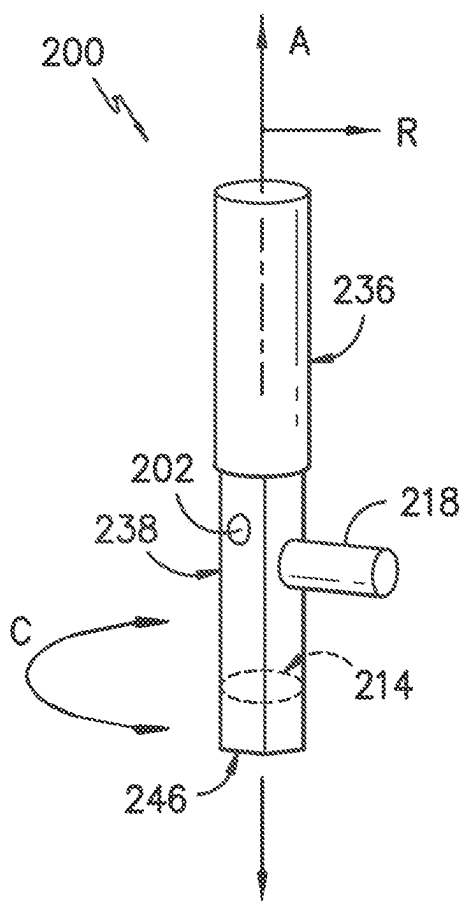


FIG. -3-



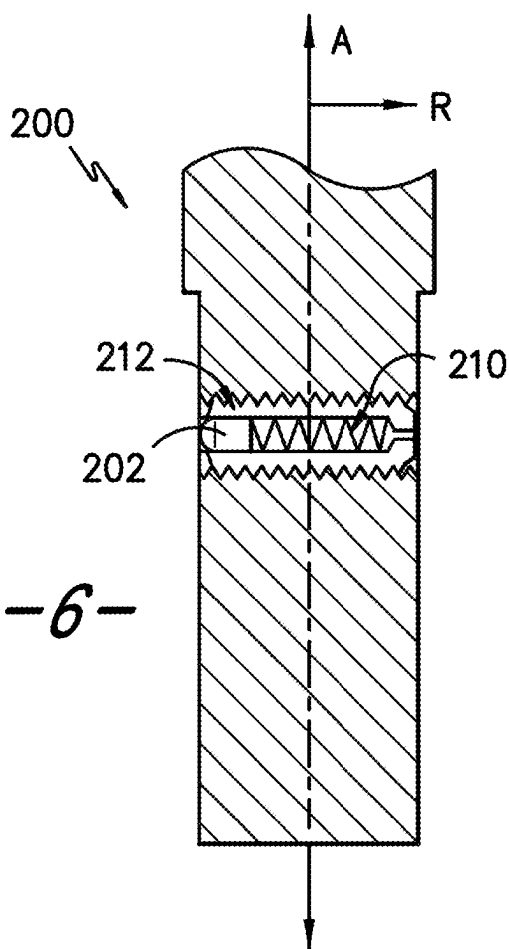


FIG. -6-

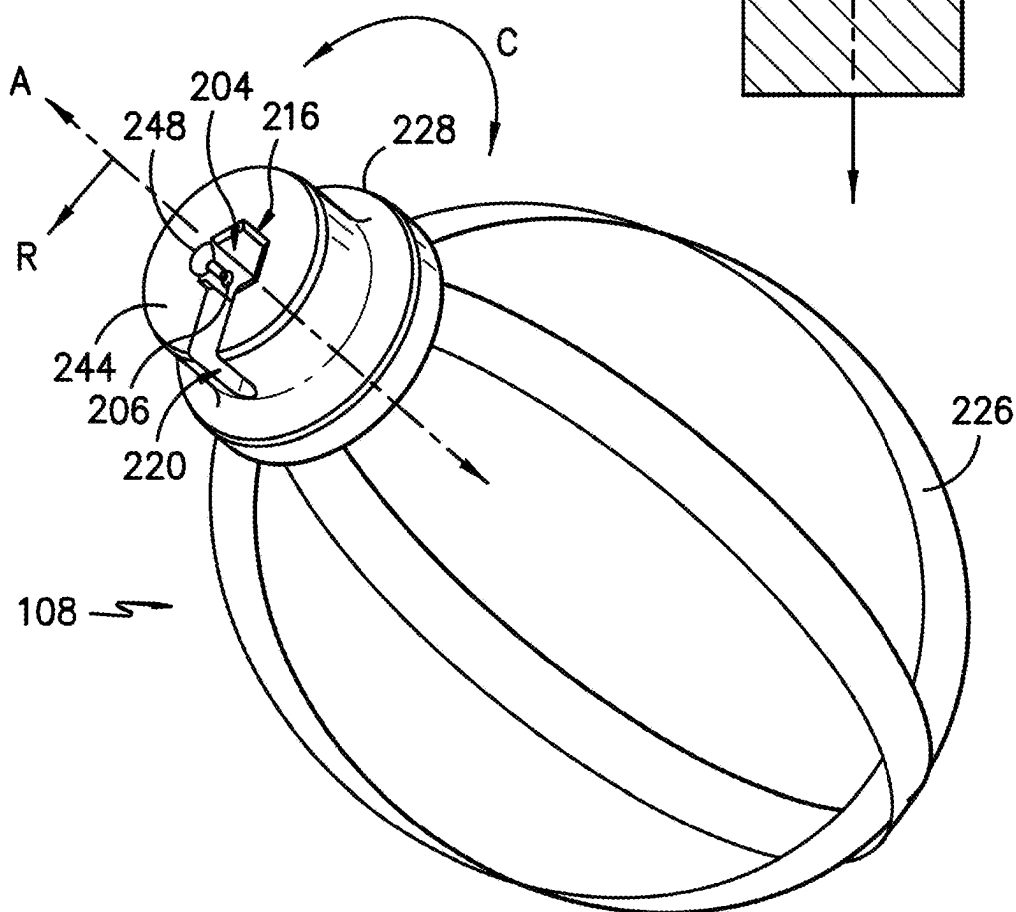


FIG. -7-

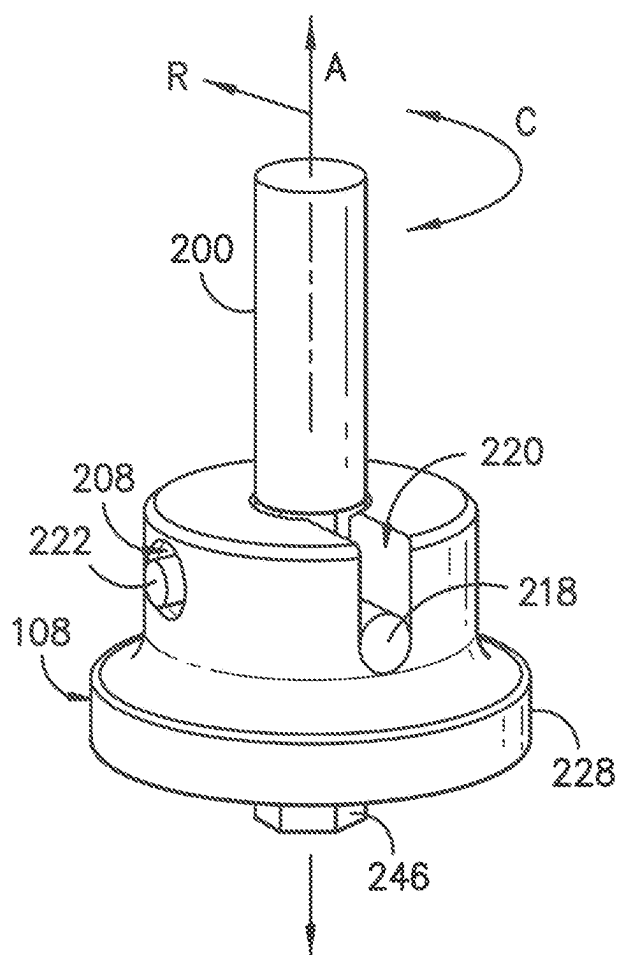


FIG. -8-

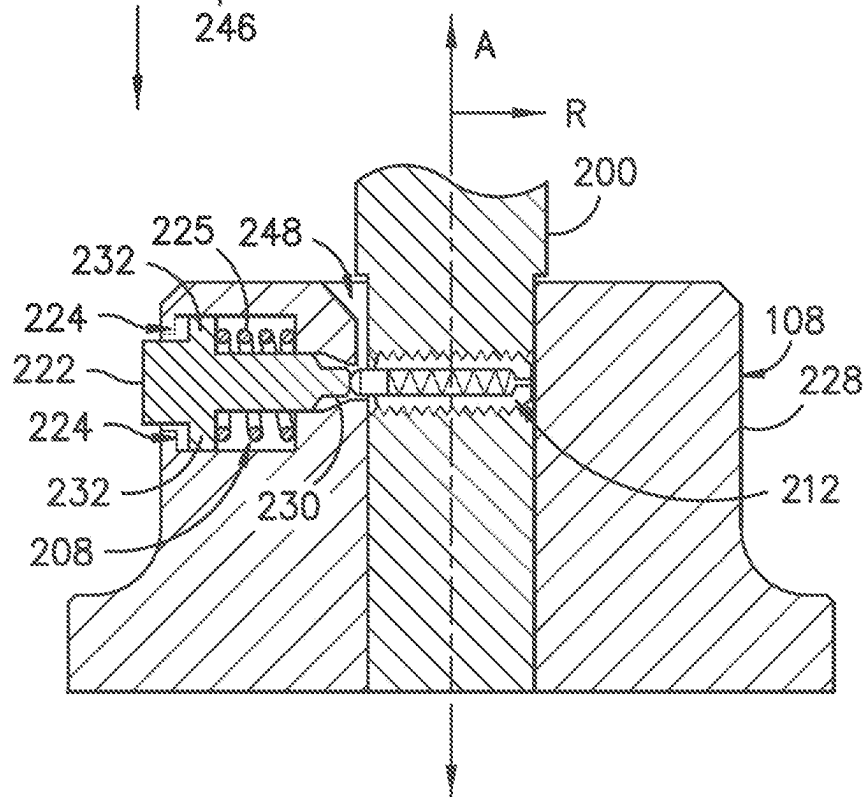


FIG. -9-

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REVERSIBLE MIXING SHAFT AND ATTACHMENT FOR STAND MIXER

FIELD OF THE INVENTION

The present subject matter relates generally to appliances, and more particularly to stand mixer appliances.

BACKGROUND OF THE INVENTION

Stand mixers generally include a stand upon which a bowl is placed and a mixing device that rotates within the bowl. In some stand mixers, the mixing device is a removable attachment that attaches to a shaft, which may be attached to the stand. The shaft may be attached to a motor located within the stand, with the motor designed to rotate the shaft in one direction. Generally, the shaft is rotatable about a shaft axis and the shaft is further rotatable about a central axis to create planetary rotation.

Mixing attachments may be exchangeable. To secure the removable mixing attachments to the shaft during rotation, various features may be used to connect each mixing attachment to the shaft. For example, a mixing attachment may have features that allow for a secure attachment when the shaft is rotated in a forward direction (e.g., clockwise). Generally, shafts with removable attachments are not run in a reverse direction (e.g., counterclockwise), as such movement would be likely to loosen or even remove the mixing attachment from the shaft during the reverse motion.

Challenges exist in the design and use of mixing attachments. Mixing attachments need to be readily removable for cleaning but also securely attached while in use. Also, certain existing constructions may utilize a design that only secures the mixing attachments while being rotated in one direction. Rotation in the opposite direction may release the mixing attachments from the shaft. This is disadvantageous because some users may prefer a mixer that can be operated in both directions.

Accordingly, a stand mixer with the ability to rotate in more than one rotation direction would be desirable. A stand mixer with removable mixing attachments that was also capable of rotating in both clockwise and counterclockwise directions would be beneficial. A stand mixer that has a shaft and removable mixing attachments that can be readily aligned and locked together would also be beneficial.

BRIEF DESCRIPTION OF THE INVENTION

Aspects and advantages of the invention will be set forth in part in the following description, or may be obvious from the description, or may be learned through practice of the invention.

In one exemplary aspect of the present disclosure, a stand mixer is provided. The stand mixer may include a motor, a mixing shaft coupled to the motor and defining an axial direction, a retractable plunger, and a mixing attachment. The mixing shaft may be rotatable by the motor about the axial direction. The retractable plunger may be extendable between a first position into the mixing shaft and a second position extending from the mixing shaft. The mixing attachment may be removably attached to the mixing shaft. The mixing attachment may define a shaft channel extending along the axial direction for sliding receipt of the mixing shaft. The mixing attachment may also define an aperture connected with the with the shaft channel and configured for selective receipt of the retractable plunger when in the second position. The mixing shaft can be inserted into the

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shaft channel and may be moved along the axial direction to selectively lock the mixing attachment to the mixing shaft by receipt of the retractable plunger into the second position and into the aperture of the mixing attachment.

In another exemplary aspect of the present disclosure, a stand mixer is provided. The stand mixer may include a motor, a mixing shaft coupled to the motor and defining an axial direction, a retractable plunger, and a mixing attachment. The mixing shaft may be rotatable by the motor about the axial direction. The retractable plunger may be extendable between a first position into the mixing shaft and a second position extending from the mixing shaft. The mixing attachment may be removably attached to the mixing shaft. The mixing attachment may define a shaft channel extending along the axial direction for sliding receipt of the mixing shaft. The mixing attachment may also define an aperture connected with the with the shaft channel and configured for selective receipt of the retractable plunger when in the second position. The mixing shaft can be inserted into the shaft channel and may be moved along the axial direction to selectively lock the mixing attachment to the mixing shaft by receipt of the retractable plunger into the second position and into the aperture of the mixing attachment. The motor may be operable to selectively rotate the mixing attachment in a clockwise direction and a counterclockwise direction around the axial direction.

These and other features, aspects and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures.

FIG. 1 provides a perspective view of an example stand mixer with a bowl according to exemplary embodiments of the present disclosure.

FIG. 2 provides a perspective view of the stand mixer with a mixing attachment according to exemplary embodiments of the present disclosure.

FIG. 3 provides a perspective view of the example stand mixer of FIG. 1 according to exemplary embodiments of the present disclosure.

FIG. 4 provides a perspective view of the example mixing shaft of FIG. 1 according to exemplary embodiments of the present disclosure.

FIG. 5 provides a cross section view of the example mixing shaft of FIG. 4 in a second position according to exemplary embodiments of the present disclosure.

FIG. 6 provides a cross section view of the example mixing shaft of FIG. 4 in a first position according to exemplary embodiments of the present disclosure.

FIG. 7 provides a perspective view of the example mixing attachment of FIG. 2 according to exemplary embodiments of the present disclosure.

FIG. 8 provides a perspective view of a portion of an example mixing shaft and a portion of an exemplary mixing attachment of FIG. 1 according to exemplary embodiments of the present disclosure.

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FIG. 9 provides a cross section view of the example mixing shaft and the exemplary mixing attachment of FIG. 8 according to exemplary embodiments of the present disclosure.

Use of the same or similar reference numerals in the figures denotes the same or similar features unless the context indicates otherwise.

DETAILED DESCRIPTION

Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

As used herein, the terms “includes” and “including” are intended to be inclusive in a manner similar to the term “comprising.” Similarly, the term “or” is generally intended to be inclusive (i.e., “A or B” is intended to mean “A or B or both”). Approximating language, as used herein throughout the specification and claims, is applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related.

The terms “coupled,” “fixed,” “attached to,” and the like refer to both direct coupling, fixing, or attaching, as well as indirect coupling, fixing, or attaching through one or more intermediate components or features, unless otherwise specified herein.

The present invention advantageously provides a mixer appliance with a secure mixing attachment coupling to a mixing shaft of the mixer appliance. This secure coupling may allow the mixer to rotate the shaft with the secured mixing attachment in clockwise and counterclockwise motions. In some embodiments, the ability to rotate mixing attachments in more than one direction may allow for faster, more complete, or more preferred mixing, blending, or other mechanical manipulation of articles mixed or otherwise manipulated by the mixer appliance.

The figures depict an exemplary stand mixer appliance 100 that may be configured in accordance with various aspects of the present disclosure. It should be appreciated that the invention is not limited to any particular style, size, model, or shape for stand mixer appliance 100. The exemplary embodiment in FIG. 1 is for illustrative purposes only. For example, appliance 100 may have different shapes and appearance for one or more parts, different motor and gear configurations, and other differences while remaining within the scope of the claimed subject matter.

With reference for FIGS. 1 and 2, for the particular embodiment shown, mixer appliance 100 includes a head 102 and a base 104 with a casing 106 surrounding head 102. Stand mixer appliance 100 extends between head 102 and base 104 in a vertical direction V, across head 102 in a lateral direction L, and from a front 103 to a back 105 in a transverse direction T. The vertical direction V, lateral direction L, and transverse direction T are perpendicular to one another.

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Head 102 is pivotally mounted to base 104 and extends transversely between front 103 and back 105 of stand mixer appliance 100 when in the mixing position shown in FIG. 1. In some embodiments, head 102 may be non-pivotally attached to base 104, but support members for bowl 98 (not shown) may raise and lower. Other configurations may be used to allow for access to bowl 98 or to a removable mixing attachment 108, as otherwise understood. For this embodiment, base 104 includes upright support 112 and a horizontal base member 116. As shown, upright support 112 extends vertically from horizontal base member 116 and horizontal base member 116 extends transversely in front of upright support 112.

Horizontal base member 116 includes a bowl seat 130 that may be rotatable about a vertical axis. In some embodiments, bowl seat 130 may be concave, grooved, or otherwise shaped to accept bowl 98. In some embodiments, support arms (not shown) or equivalent support structures may be used to hold bowl 98, as would be understood. Other shapes and constructions may be used for bowl stand 120 as well.

Head 102 includes an attachment support 110. A motor 142 is disposed within head 102 and enclosed by casing 106. Attachment support 110 is located on a lower portion or underside 126 of head 102 and forward of upright support 112 along transverse direction T. A mixing shaft 200 extends from attachment support 110. Removable mixing attachment 108 removably attaches to shaft 200.

Drivetrain 144 connects motor 142 with one or more gears 146 for causing rotation of attachment 108 or mixing shaft 200. Gears 146 may allow for selection by the user of different rotating speeds for attachment 108. As will be described in more detail below, mixing attachment 108 may be locked onto shaft 200 prior to rotation of shaft 200 by motor 142.

Appliance 100 may include one or more controls 150 for operations such as selectively powering motor 142, choosing the speed of rotation for attachments 108, locking position of head 102 relative to base 104 during mixing, or other features. In some embodiments, controls 150 may include a rotational direction operation selection, allowing a user to select the direction of rotation of the mixing shaft 200.

In certain embodiments, attachment support 110 may accept more than one attachment 108. Various types of attachments 108 may be used including e.g., whisks, paddles, dough hooks, beaters, and others for purposes of mixing articles or mechanically manipulating articles within bowl 98 or other containers supported by base 104. During use, attachment support 110 with mixing shaft 200 may rotate attachment 108 in a circular or planetary fashion. Spinning in a planetary fashion, as used herein, includes spinning an object (e.g., shaft 200) about a first axis and spinning the object about a second axis, the object offset from the second axis. For example, shaft 200 may spin about a shaft axis SA, and spin about a central axis CA, shaft 200 offset from central axis CA to generate spinning in a planetary rotation. Shaft axis SA may also be offset from central axis CA. In some embodiments, motor 142 may be disposed within base 104, including within upright support 112.

As shown in FIG. 2, mixing shaft 200 may rotate within attachment support 110. Locked mixing attachment 108 and mixing shaft 200 are rotatable by motor 142 in planetary rotation. Mixing shaft may define an axial direction A, with a radial direction R extending therefrom, and a circumferential direction C extending around axial direction (as shown in FIG. 4). Mixing shaft 200 may rotate around axial

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direction A, mixing shaft **200** further rotating in circumferential direction C in addition to axial direction A. According to the arrangement, axial direction A may correspond to shaft axis SA in some embodiments. Additionally or alternatively, motor **142** may be operable to selectively rotate mixing attachment **108** in a clockwise direction or a counterclockwise direction in circumferential direction C around axial direction A. Mixing shaft **200** may define axial direction A, with radial direction R extending therefrom, and circumferential direction C defined around mixing shaft **200**. Thus, mixing shaft **200** may be reversible, or moveable in either direction during use. Attachment of mixing attachment **108** to shaft **200** allows for motion in both directions, clockwise and counterclockwise, by motor **142**. In other words, motor **142** can rotate mixing attachment **108** or shaft **200** in a clockwise direction and can switch and rotate mixing attachment **108** or shaft **200** in a counterclockwise direction. Such movement may be directed by a user (e.g., by use of controls **150**) or may be directed independent of a user, e.g., by using a timer, by using a controller in operable communication with motor **142**, or as otherwise understood.

Turning generally to FIGS. **3** through **9**, FIG. **3** depicts mixer appliance **100** with mixing attachment **108** and bowl **98** removed. FIG. **4** depicts a partial perspective view of shaft **200**. As shown, mixing shaft **200** defines a plunger chamber **212** (as shown in FIG. **5**) and includes a retractable plunger **202**. Retractable plunger **202** may be contained within plunger chamber **212**, as will be described in more detail below. Mixing shaft **200** may also include a resilient element (e.g., spring **210**) connected to plunger **202** (e.g., as shown in FIGS. **5** and **6**). Resilient element (e.g., spring **210**) may additionally be a hydraulic system, an air compression system, or an equivalent structure. In some embodiments, mixing shaft **200** includes an alignment rod **218**, which will be described in more detail below. In some embodiments, mixing shaft **200** further includes an upper portion **236** and a lower portion **238**. Mixing shaft **200** is a rod extending vertically below attachment support **110**. Mixing shaft **200** is coupled to motor **142**. Coupling to motor **142** may be a direct coupling or may be an indirect coupling, such as through drivetrain **144** or gears **146**.

Upper portion **236** is generally above lower portion **238** along the vertical direction V (e.g., which may correspond to the axial direction A). In some embodiments, mixing shaft **200** has a different shape between upper portion **236** and lower portion **238**. For example as shown, upper portion **236** has a cylindrical shape with a circular cross-sectional shape and lower portion **238** has a noncircular cross-sectional shape. As shown in FIG. **4**, lower portion **238** may have a hexagonal cross-sectional shape **214** in a plane orthogonal to axial direction A. In some embodiments, upper portion **236** may be inside attachment support **110** or inside head **102** of mixer appliance **100**. As shown in FIG. **3**, in some embodiments, upper portion **236** extends below (in axial direction A or in vertical direction V) attachment support **110** or head **102**. In certain embodiments, shaft **200** may have noncircular cross-sectional shape, such as hexagonal cross-sectional shape **214**, along a majority or along the entirety of shaft **200**. In some embodiments, noncircular cross-sectional shape of shaft **200** has at least three corners or at least three sides.

As shown in FIGS. **3** and **4**, alignment rod **218** extends from mixing shaft **200** along radial direction R. Alignment rod **218** may be any shape, such as a cylindrical shape or a cubic shape. In some embodiments, Alignment rod **218** is

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offset in circumferential direction C from retractable plunger **202**, which resides in plunger chamber **212** (as shown in FIG. **5**).

Retractable plunger **202** is moveable in a radial direction R. FIG. **5** shows retractable plunger **202** in a second position or an extended position. FIG. **6** shows retractable plunger **202** in a first position or a retracted position. Retractable plunger **202** is extendable along radial direction R from mixing shaft **200**. Plunger is extendable between first position into mixing shaft **200** and second position extending from mixing shaft **200**. In first position, retractable plunger **202** may be flush or mostly flush with shaft in radial direction R. Spring **210** may be compressed around plunger **202** in first position. In second position, plunger **202** may extend in radial direction R from shaft **200**. Spring **210** is relaxed around plunger **202** in second position. Generally, spring **210** is positioned to urge plunger **202** toward second position.

In some embodiments, retractable plunger **202**, spring **210** and plunger chamber **212** may be a plunger sub-assembly **240**, plunger sub-assembly **240** insertable within mixing shaft **200** within a sub-assembly cavity **242**, as shown in FIG. **5**. Plunger sub-assembly **240** may be inserted into mixing shaft **200** by compression fitting, screw grooves and flanges, adhesives, or as otherwise understood. Spring **210** and plunger **202** may be contained in plunger sub-assembly **240** by screwing means, compression fitting, adhesives, or as otherwise understood.

Turning to FIG. **7**, which shows mixing attachment **108**, mixing attachment **108** includes an attachment head **228** and at least one mixing apparatus (shown as a plurality of paddles **226** in FIG. **7**). Mixing attachment **108** further defines a shaft channel **204** for sliding receipt of mixing shaft **200**. Shaft channel **204** extends along axial direction A when mixing attachment **108** is attached to shaft **200**. As shown, shaft channel **204** extends through a central portion of attachment head **228**. In some embodiments, shaft channel **204** may extend through a portion of attachment head **228** or through all of attachment head **228**. Additionally or alternatively, mixing attachment **108** may include an attachment top **244**. As shown, attachment top **244** is located axially above the rest of mixing attachment **108**.

Generally, shaft channel **204**'s noncircular cross-sectional shape (e.g., hexagonal cross-sectional shaft channel shape **216**) corresponds to the shape of shaft **200**'s cross-sectional shape (e.g., hexagonal cross-sectional shape **214**). In certain embodiments, shaft channel **204** may have noncircular cross-sectional shape, such as hexagonal cross-sectional shaft channel shape **216**, along shaft channel **204** in axial direction A. In other words, mixing shaft **200** may include hexagonal cross-sectional shaft channel shape **216** in a plane orthogonal to axial direction A. In some embodiments, the noncircular cross-sectional shape of shaft channel **204** may have at least three corners or at least three sides. The matching shape of shaft channel **204** to shaft **200** may prevent or decrease movement of mixing attachment **108** relative to shaft **200** during movement of shaft **200** relative to head **102** of mixer appliance **100** by motor **142**. As shown in FIG. **7**, shaft channel **204** may form hexagonal cross-sectional shaft channel shape **216** that corresponds to cross-sectional shape **214** of shaft **200**. During use, mixing attachment **108** is prevented from rotating relative to mixing shaft **200** when mixing attachment **108** is attached to mixing shaft **200**. The corners and edges of shaft **200** and shaft channel **204** may secure or aid in securing mixing attachment **108** to shaft **200** in circumferential direction C. Advantageously, secure attachment of mixing attachment **108** to shaft **200**

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may allow motor **142** to rotate shaft **200** with secured mixing attachment **108** in either a clockwise or a counterclockwise direction. In other words, motor **142** may switch between a clockwise and a counterclockwise rotation of shaft **200** (e.g., including planetary rotational patterns, and axial rotation or rotation about axial direction A), with mixing attachment **108** remaining securely attached to shaft **200** during both directions of rotation.

As shown, mixing attachment **108** further defines an aperture **206** connected with shaft channel **204**. Aperture **206** may be configured for selective receipt of retractable plunger **202** when in second position (i.e., plunger **202** extended from shaft **200** in radial direction R). Aperture **206** may be sized to allow retractable plunger **202** to extend into second position when in aperture **206**.

Additionally or alternatively, mixing attachment **108** also defines a plunger track **248** extending along axial direction A adjacent to shaft channel **204**. During use, plunger track **248** may be for sliding receipt of plunger **202** as mixing shaft **200** moves along shaft channel **204**, plunger track **248** connected with aperture **206**. As shown in FIGS. 7 and 9, plunger track **248** extends radially into mixing attachment **108** from shaft channel **204**. Plunger track **248** may also extend along shaft channel **204** in axial direction A, widening in a partial conical shape upwards towards attachment support **110** and connecting with aperture **206** downwards into shaft channel **204** towards mixing paddles **226** or towards bowl **98**. Plunger track **248** radially extends from shaft channel **204** in radial direction R, with a conical shape that is widest at attachment top **244**. In some embodiments, plunger track **248** may continue from the point of the conical shape into a cylindrical or half pipe shape as it extends towards aperture **206**. Plunger track **248** then decreases in a length in circumferential direction C axially downward, connecting to aperture **206**.

FIG. 8 depicts mixing attachment **108** (e.g., attachment head **228**) attached to shaft **200**. Aperture **206** may be sized to allow retractable plunger **202** to extend into second position when in aperture **206**. In some embodiments, aperture **206** is deeper than surrounding plunger track **248** (e.g., as seen in FIG. 9) in radial direction R. In some embodiments, mixing shaft **200** can be inserted into shaft channel **204** (e.g., at bottom end of shaft **246**) and moved along axial direction A to selectively lock mixing attachment **108** to mixing shaft **200** by receipt of retractable plunger **202** into second position and into aperture **206** of mixing attachment **108**. As shown, placing plunger **202** into aperture **206** locks mixing attachment **108** to shaft **200**. According to the arrangement, plunger **202** locked into aperture **206** aids in restricting movement of mixing attachment **108** in relation to shaft **200**, for example, in axial direction A.

Optionally, mixing attachment **108** defines a rod channel **220**. As shown in FIGS. 7 and 8, rod channel **220** may extend radially out from shaft channel **204**. In some embodiments, rod channel **220** may extend along axial direction A. For example, rod channel **220** may extend downwards from attachment top **244**. Generally, rod channel **220** is configured for sliding receipt of alignment rod **218**. Rod channel **220** is further sized to allow alignment rod **218** to fit into rod channel **220**. During use, rod channel **220** may be aligned with alignment rod **218** to aid in attaching mixing attachment **108** to shaft **200**. Alignment rod **218** may be configured to align plunger track **248** of mixing attachment **108** with retractable plunger **202**. Seating alignment rod **218** into rod channel **220** may align retractable plunger **202** with aperture **206**, securing mixing attachment **108** onto mixing shaft **200**. As shown, rod channel **220** is offset (e.g., in circumferential

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direction C) from retractable plunger **202** on attachment head **228** of mixing attachment **108**.

As shown in FIGS. 8 and 9, stand mixer appliance **100** further includes a release button **222** and a chamber **208** defined by mixing attachment **108**. FIG. 9 depicts a cross sectional view of shaft **200** and mixing attachment **108**, including chamber **208** and release button **222**. Chamber **208** for release button **222** is located adjacent to aperture **206**. As shown, chamber **208** may be farther from shaft channel **204** in radial direction R than aperture **206**. In some embodiments, chamber **208** is connected to aperture **206** or to plunger track **248**. As shown, chamber **208** may house release button **222** within it. Release button **222** further extends from chamber **208** in radial direction R in some embodiments.

Release button **222** positioned in chamber **208** is moveable along radial direction R towards and away from aperture **206**. As shown in FIG. 9, release button **222** may contact plunger **202** in second position and mixing attachment **108** is secured on mixing shaft **200**. In other words, release button **222** may contact plunger **202** when plunger **202** is locked in aperture **206**. During use, release button **222** may push plunger **202** from second position towards first position. Accordingly, release button **222** may unlock plunger **202** from aperture **206**, allowing shaft **200** and mixing attachment **108** to disconnect from one another. As shown in FIG. 9, release button **222** extends along radial direction R from mixing shaft **200**.

In some embodiments, release button **222** includes button flanges **232**, located axially above or below a main body of release button **222**. During use, button flanges **232** guide release button **222** in radial direction R along chamber **208** to compress retractable plunger **202** into first position.

In some embodiments, at least one retention flange **224** may extend around chamber **208** to hold release button **222** in chamber **208**. Retention flanges **224** may extend around an opening to chamber **208** or may extend as tabs extending into chamber **208** (e.g., as shown as opposing retention flanges **224** in FIG. 9). In certain embodiments, a button spring **225** coils around release button **222**. Button spring **225** may bias release button **222** outward from shaft **200** in radial direction R, locking mixing attachment **108** onto shaft **200** unless release button **222** is compressed toward plunger **202** in radial direction R. In some embodiments, button spring **225** may be a compression spring. In some embodiments, button spring **225** may restrain release button **222** in chamber **208**. Additionally or alternatively, equivalent mechanisms may restrain release button **222** in chamber **208** or bias release button **222** in radial direction R away from shaft **200**. For example, release button **220** may be retained within chamber **208** by a circlip, a snap ring, or a press foot ring.

In some embodiments, release button **222** includes button end **230**. Button end **230** may be in physical contact with retractable plunger **202**. In some embodiments, button end **230** is only in contact with retractable plunger **202** during movement of button **222** to compress retractable plunger **202** into first position. In some embodiments, and as shown in FIG. 9, retractable plunger **202** is in contact with button end **230** when mixing attachment **108** is attached to shaft **200**, whether retractable plunger **202** is locked in second position in aperture **206** or retractable plunger **202** is unlocked and not in second position (e.g., when plunger **202** is compressed into first position).

Additionally or alternatively, chamber **208** may be an empty chamber in some embodiments. A pin, rod, or other equivalent structure may be inserted into chamber **208** to

release mixing attachment **108** from shaft **200**, pressing plunger **202** into first position to release mixing attachment **108** from shaft **200**.

Aspects of the present subject matter are directed to a method and system to enable reversibility on a stand mixer attachment that may add ease of use to loading and unloading stand mixer attachments, may provide strong and secure locking of any attachment, and may allow all attachments to be used in forward and reverse. The system may have a locking mechanism on a stand mixer that uses a hex end motor shaft which encompasses a spring-loaded plunger within the shaft. The attachment head may have a ramped top and guide channel which the spring-loaded plunger may fall into and may be pushed back. When the attachment is slid further onto the shaft the compressed spring plunger may fall into a hole on the attachment top. This may prevent the attachment from moving up or down and the hex shaft may prevent the attachment from rolling side to side. There may further be a stepped push button on the side of the attachment as a release. When the button is pushed it may press the spring plunger in, allowing the attachment to be released from the motor shaft.

This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

What is claimed is:

1. A stand mixer, comprising:
 - a motor;
 - a mixing shaft coupled to the motor and defining an axial direction, the mixing shaft rotatable by the motor about the axial direction;
 - a retractable plunger extendable along a radial direction from the mixing shaft, the retractable plunger extendable between a first position into the mixing shaft and a second position extending from the mixing shaft;
 - a mixing attachment removably attached to the mixing shaft, the mixing attachment defining a shaft channel extending along the axial direction for sliding receipt of the mixing shaft, the mixing attachment also defining an aperture connected with the shaft channel and configured for selective receipt of the retractable plunger when in the second position,
 wherein the mixing shaft can be inserted into the shaft channel and moved along the axial direction to selectively lock the mixing attachment to the mixing shaft by receipt of the retractable plunger into the second position and into the aperture of the mixing attachment.
2. The stand mixer of claim 1, wherein the motor is operable to selectively rotate the mixing attachment in a clockwise direction and a counterclockwise direction around the axial direction.
3. The stand mixer of claim 1, wherein the mixing shaft further comprises a resilient element, and
 - wherein the mixing shaft defines a plunger chamber containing the retractable plunger and the resilient element, the resilient element positioned to urge the retractable plunger towards the second position.

4. The stand mixer of claim 1, the mixing attachment further comprising a plunger track extending along the axial direction adjacent to the shaft channel, the plunger track configured for sliding receipt of the retractable plunger as the mixing shaft moves along the shaft channel, the plunger track connected with the aperture.

5. The stand mixer of claim 1, wherein the mixing shaft comprises a noncircular cross-sectional shape and the shaft channel comprises a complementary noncircular cross-sectional shape such that the mixing attachment is prevented from rotating relative to the mixing shaft when the mixing attachment is attached to the mixing shaft.

6. The stand mixer of claim 5, wherein the mixing shaft further comprises a hexagonal cross-sectional shape in a plane orthogonal to the axial direction.

7. The stand mixer of claim 1, further comprising a plunger track extending along the axial direction adjacent to the shaft channel of the mixing attachment, the plunger track configured for sliding receipt of the retractable plunger as the mixing shaft moves along the shaft channel;

an alignment rod extending from the mixing shaft along the radial direction; and

a rod channel defined by the mixing attachment, the rod channel extending along the axial direction and configured for sliding receipt of the alignment rod, wherein the alignment rod is configured to align the plunger track of the mixing attachment with the retractable plunger.

8. The stand mixer of claim 1, further comprising a release button;

a chamber defined by the mixing attachment and located adjacent to the aperture, the release button positioned in the chamber and movable along the radial direction towards and away from the aperture; and

whereby the release button is configured to push the retractable plunger from the second position towards the first position so as to allow the mixing attachment to be removed from the mixing shaft.

9. The stand mixer of claim 8, wherein the release button is retained within the chamber, and oriented at an opening of the chamber by one of: a retention flange, a circlip, a snap ring, and a press foot ring.

10. The stand mixer of claim 9, wherein the release button contacts the retractable plunger when in the second position and the mixing attachment is secured on the mixing shaft.

11. A stand mixer comprising:

a motor;

an attachment support coupled to the motor;

a mixing shaft coupled to the motor and to the attachment support and defining an axial direction, the mixing shaft rotatable by the motor about the axial direction;

a retractable plunger extendable along a radial direction from the mixing shaft, the retractable plunger extendable between a first position in the mixing shaft and a second position extending from the mixing shaft; and

a mixing attachment removably attached to the mixing shaft, the mixing attachment defining a shaft channel extending along the axial direction for sliding receipt of the mixing shaft, the mixing attachment also defining an aperture connected with the shaft channel for selective receipt of the retractable plunger when in the second position,

wherein the mixing shaft can be inserted into the shaft channel and moved along the axial direction to selectively lock the mixing attachment to the mixing shaft

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by receipt of the retractable plunger into the second position and into the aperture of the mixing attachment, and

wherein the motor is operable to selectively rotate the mixing attachment in a clockwise direction and a counterclockwise direction around the axial direction.

12. The stand mixer of claim **11**, wherein the mixing shaft is rotatable by the motor in a planetary rotation, the mixing shaft rotating around a shaft axis extending along the mixing shaft, and the attachment support spinning about a central axis, the central axis extending in the axial direction.

13. The stand mixer of claim **11**, wherein the mixing shaft comprises a chamber containing the retractable plunger and a spring, the spring positioned to urge the retractable plunger towards the second position.

14. The stand mixer of claim **11**, further comprising a plunger track extending along the axial direction adjacent to the shaft channel, the plunger track configured for sliding receipt of the retractable plunger as the mixing shaft moves along the shaft channel, the plunger track connected with the aperture.

15. The stand mixer of claim **11**, wherein the mixing shaft comprises a noncircular cross-sectional shape and the shaft channel comprises a corresponding noncircular cross-sectional shape such that the mixing attachment is prevented from rotating relative to the mixing shaft when the mixing attachment is attached to the mixing shaft.

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16. The stand mixer of claim **11**, wherein the mixing shaft further comprises a hexagonal cross-sectional shape in a plane orthogonal to the axial direction.

17. The stand mixer of claim **11**, further comprising a plunger track extending along the axial direction adjacent to the shaft channel, the plunger track configured for sliding receipt of the retractable plunger as the mixing shaft moves along the shaft channel;

an alignment rod extending from the mixing shaft along the radial direction; and

a rod channel defined by the mixing attachment, the rod channel extending along the axial direction and configured for sliding receipt of the alignment rod, wherein the alignment rod is configured to align the plunger track of the mixing attachment with the retractable plunger.

18. The stand mixer of claim **11**, further comprising a release button;

a chamber defined by the mixing attachment and located adjacent to the aperture, the release button positioned in the chamber and movable along the radial direction towards and away from the aperture; and

whereby the release button is configured to push the retractable plunger from the second position towards the first position so as to allow the mixing attachment to be removed from the mixing shaft, and

wherein the release button extends along the radial direction from the mixing shaft.

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